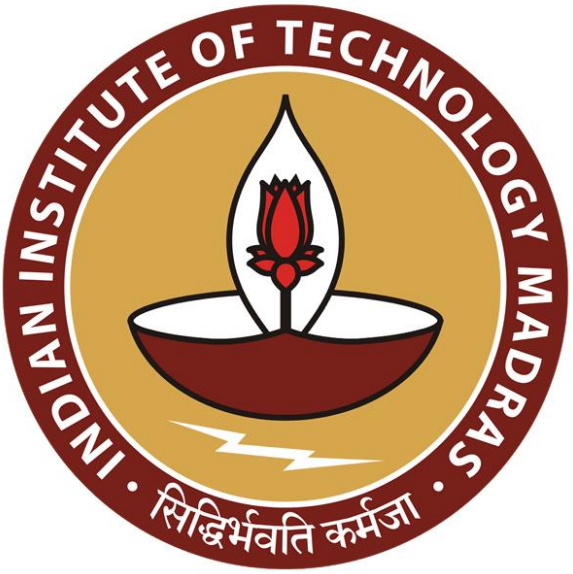
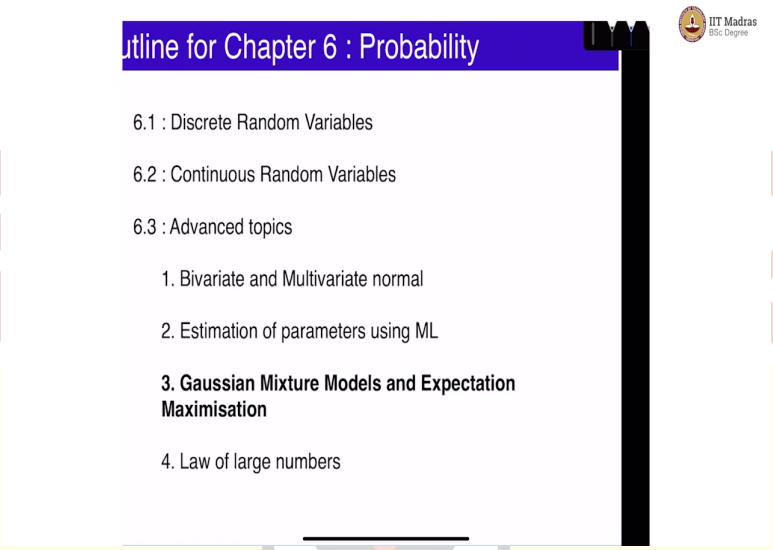


IIT Madras

ONLINE DEGREE

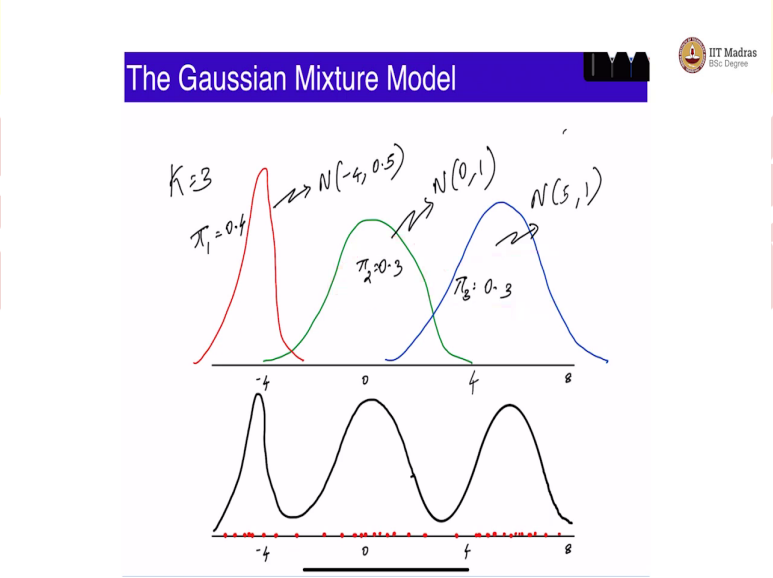
Machine Learning Foundations
Professor Harish Guruprasad Ramaswamy
Department of Computer Science and Engineering
Indian Institute of Technology, Madras
Gaussian Mixture Models Expectation and Maximisation

(Refer Slide Time: 0:13)



Now, let us consider the next topic, which is the Gaussian mixture models and the expectation maximization algorithm.

(Refer Slide Time: 0:22)



Before we define what Gaussian mixture model is, I will first give a motivating example. Let us say you have an exam and there are three types of students who write the exam. And there is one type of student who mostly it is a multiple choice exam, let us say, and who mostly

chooses option A, and occasionally chooses option B, C, or D. So, there is one type of student who does not study and who mostly chooses option A because he prefers it and sometimes it he chooses B, C, and D.

And then there is another type of student who just randomly chooses A, B, C, D. He rolls a 4 sided die and decides A, B, C, D at the random. And then there is a third type of student who studies and prepares and writes the answers. So, there are three types of students. And let us say this is an exam with negative marks, and so on.

And let us say the question paper is such that the student who mostly prefers option A would, on average, get a negative - 4 as his marks with the variance given by let us say, 0.5, so that is the marks achieved, the score achieved by the student who mostly chooses option A and does chooses option B, C, D, at random for some other fraction of the time. Let us say the score of that particular type of student is a normal random variable with mean - 4 and variance 0.5.

So, let us assume that that is the red curve and we say that the second type of student who chooses 4 options uniformly at random. And let us say the score achieved by such a student is normal with mean 0 and variance 1, we will say that it is given by the green curve. And then finally, you have the student who studies and prepares and thinks and answers. And the, the score achieved by such a student is, let us say, a normal random variable with mean given by 5 and variance given by 1.

This is just an example here, and let us say that is given by the blue curve. That is why I have the red curve corresponds to normal with mean - 4 and variance 0.5, the green curve corresponds to normal with mean 0, variance 1 and the blue curve corresponds to normal with mean 5 and variance 1. And let us say 40 percentage of students belong to the first type, 30 percent belong to the second type, and another 30 percent belong to the third type. That is what is captured by this three numbers.

This is just to summarize the case here, there are three types of students and each, the marks received by each student is a normal given by these things. Of course, this is not, let us say, this is not something which someone advertises what type of student they are, is not known in advance what the, let us say, the grade or the teacher who corrects the papers, he just write; he or she just writes down the marks in a long column and does a histogram of the marks.

Then she will observe some strange property, that is, the marks will have three peaks there are, the marks will either be around - 4, or will be around 0, or will be around 5. So, that is

the kind of behavior that the teacher will observe. The marks seem to cluster into three types of groups, the marks is either near - 4 or near 0 or near 5. So, if the samples might look like this.

So, each of the red dots correspond to a sample point, so that is an example of how the data will look like you might get something like - 5 -, - 5.2, - 4.8, - 4 or something like that, that each of these points correspond to one data point. You can see that there is a lot of points near - 4, there is very little points in this region between - 4 and 0. There is a lot of points, a lot of people who have scored near 0 and there is very little people who have scored between in this region and there are a lot of people who have scored around 5.

If you did the histogram, if you had say 1000 or 10,000 such students and if the teacher did the histogram, the histogram would look like this. This is clearly a multi modal distribution. So, there are three peaks here. Such a distribution cannot be captured by any of the distributions we discovered, discuss till now we have discovered something discuss something like let us say, the normal distribution, the exponential distribution and so on. All of these have a single peak.

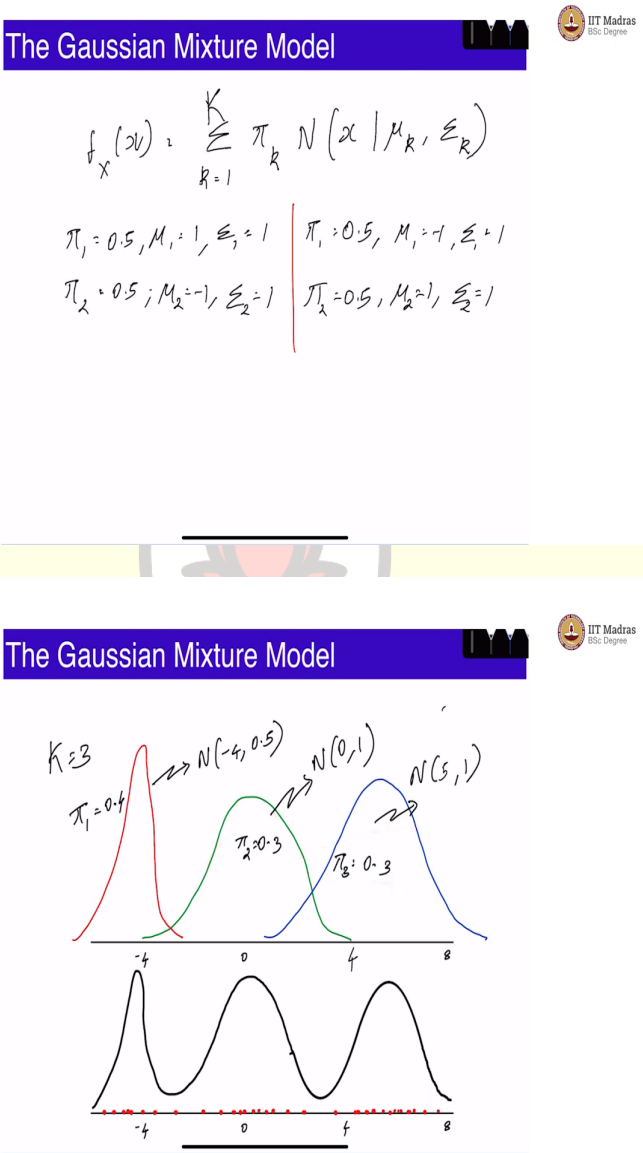
So, if you did the density, the density there was a single peak. But in a very practical example, like what we discuss right now, we have data whose density seems to have three peaks, you cannot capture such data using at least you cannot capture such data well, using any of the distributions that we have discussed till now. And this is a very natural kind of data where there is some kind of clustering that happens in the data and you want to capture this.

The right model of choice for this kind of data is what is called the Gaussian mixture model. As it says it is a mixture of Gaussians, so that is our normal distributions, it is a mixture of Gaussian or normal distributions. Let us take the same example. If the density was really like this, if there was 40 percent students who followed the red curve, 30 percent who followed the green curve and 30 percent who followed the blue curve, the total, the effective density will effectively be just a combination of the red curve, the green curve and the blue curve.

It will not be just addition, because this should also integrate to 1, the area under this curve, area under the red curve is going to be 1, the area under the green curve is going to be 1, area under the blue curve is going to be 1. But if you just add these three, the area is not going to be 1, area is going to be 3. So, that is the reason why you the right form of the density of this mixture model is simply 0.4 times the red curve, plus 0.3 times the green curve, plus 0.3 times the blue curve.

That kind of also captures which how what fraction of students are in the red mode, what fraction of students are in the green mode, what fraction of students are in the blue mode, and so on. So, we will put this together and get the Gaussian mixture model.

(Refer Slide Time: 6:49)



The Gaussian mixture model, the expression is as follows. fX of x is equal to you have let us say k components. In this case, we have 3 components. But let us say in general, you could have k components, each of these is called a component. And we will use small k as a dummy variable, and capital K is the number of components, small k equal to k1 to capital K. Pi K this is normal, evaluated at x with mean given by mu k, and covariance given by sigma k.

I have just taken the general D-dimensional case directly because there is no nothing special about one-dimension, one-dimension is easy for visualization. But other than that, there is nothing special about one-dimension. So, this is the density of the Gaussian mixture model. The parameters of this model are given by you have k, you have k components, capital K components, let us write down the capital K clearly. You have capital K components, and each component has kind of three parameters.

Pi k is a scalar parameter representing which how what fraction of the data is from component K. And you have the mean of component K and the covariance matrix of component. So, these form the parameters of the Gaussian mixture model. I have just written N here, if you want, you can expand N of x mu k sigma k using the standard multivariate normal which would simply be 1 by square, 1 by 2 pi e power d by 2 square root of determinant of sigma k x of - half into x - mu k transpose sigma k inverse x - mu kx. So that is a standard multivariate mean. That is the multivariate normal expression that we have seen before. You can write that down later if you want.

So, this is the, density of the Gaussian mixture model. This, note that this is different from the density of addition of normal mean you can have three, let us say you can have three normal distinct normal, discrete random variables and you add them, that is not the same as adding the densities. So, adding the random variables is not the same as adding the densities.

In fact, we know that if we add normal random variables, we get back normal you get another normal random if you have seen that in the properties of the normal distribution, if you add the normal random variables, you get another normal random variable, but here by adding normal densities, you get the Gaussian mixture model. I already told you this is useful for handling data, which seems to have multiple components.

So, there are different groups within the data which is not visible to you. So, Gaussian mixture model is a very good fit for data. In example, data would be what I told you before, for there are three types of students. You do not know the types of the students. They are just three types of students and the data clusters into three groups.

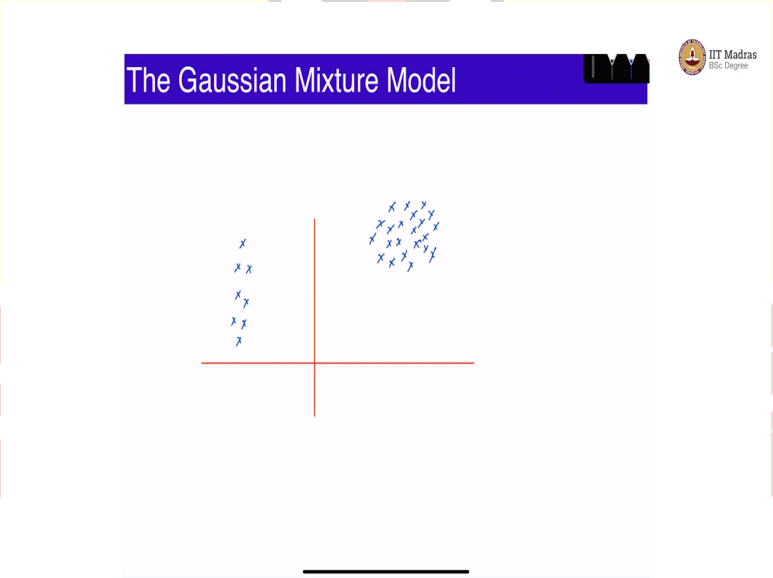
Firstly, you should note that that there is an intrinsic unidentified unidentifiability here, that is, you can always swap the components, for example, you can have two sets of parameters, let us say you have just a two component model. Let us say you can have one set of parameters, let us say pi 1 is 0.5, mu 1 is let us say 1, sigma 1 is let us say 1, pi 2 is 0.5, mu 2

is let us say - 1, and sigma 2 is let us say 1. So, this is one parameter, one set of parameters for a two component Gaussian mixture model.

Let me give another set of parameters. Let us have that let us say pi 1 is 0.5, mu 1 is - 1 sigma 1 is 1, pi 2 is 0.5, mu 2 is 1, and sigma 2 is 1. So, technically, this parameter seems to be different from this parameter. Because you can see the mu 1 is 1 and mu 2 is - 1. Here mu 1 is - 1 and mu 2 is 1. But both the set of parameters represent the same density, why is that because it is just addition. So, there is no way you can distinguish between this parameter and this parameter.

So, the indexing is kind of meaningless, just that you must understand the indexing 1 and 2, you just need to be consistent. That is the only thing here. Even though these two parameters are different, when you write down as a vector, they really represent the same density. So, that is one thing you must be aware of. So, how data from, let us say a two component Gaussian mixture model with two different means and two different variances in let us say, two dimensional data look like? So, that is a simple example.

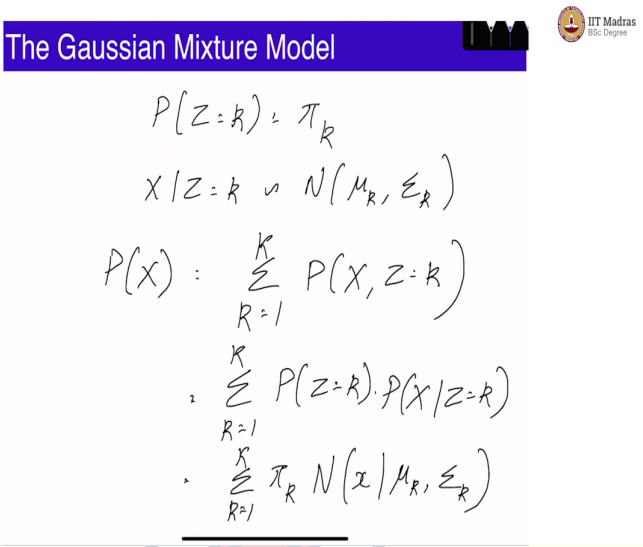
(Refer Slide Time: 12:08)



Here is a simple example of how it would look like for the two-dimensional case with two different means and two different covariances, as you can see that there are two types of data, obviously, of course, you do not have the access to the color, the color will really be the same. So, this is how it would look like for you. If you had a two dimensional data with two different means and two different covariances and you had, let us say, some 50 samples from

it, this is how this is what you would get. If you did a scatter plot of the points, you might get something like this. So, this is how data from a Gaussian mixture model would look like.

(Refer Slide Time: 12:40)



Here is an alternate interpretation of, or not alternate interpretation, it is really the same interpretation, but just that alternate notation to write the Gaussian mixture model. The one thing that we do is to introduce the latent into the latent variable z. So, in this case, in the in the case of the exams score, this latent variables Z corresponds to which type of student you are.

So, there are three types of student, are you the type 1 student, type 2 student or type 3 student is given by the hidden variable z. So, we can have Z probability of Z equal to k is given by pi k. So, that is a probability of you being a type 1 student is 0.4, being a type 2 student is 0.3, type 3 student is 0.3 and so on. So, 0.4, 0.3, 0.3 is captured by this pi k. So, this in that example that we discussed recently.

And what we have is that x given z equal to k is distributed as normal, with mean mu k, and covariance sigma k. This is again, the same thing that we discussed just that we have given a name to the variable, which represents what group of should what group do you belong to. I have just given a name for that, that is it. So, what is the joint density or what is the density of X?

I am just going to use probabilities and densities interchangeably. Just this is not strictly speaking right. But it does not matter very much this level. So, what is the probability of X? Well, it is simply you can sum over Z equal to 1 to capital K. probability of X , z is equal to

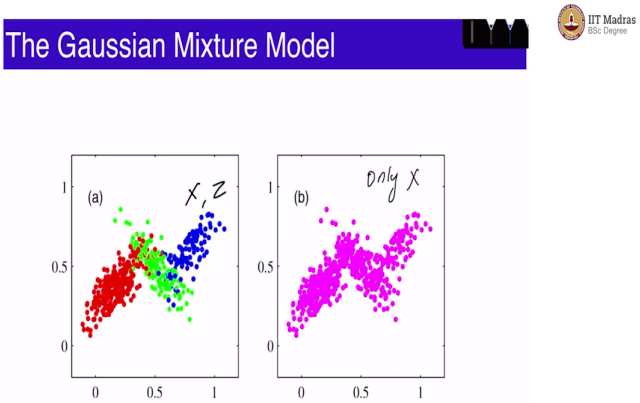
that k equal to 1 capital K probability of z equals x , z equal to k. This is always true, because I need the marginal for X. I am just taking the joint of x , Z and I am just marginalizing out Z. This is always valid.

Why am I doing that? Because I can write the joint of x , Z as x given Z into Z. That is the main advantage of the joint of x , Z which is going to be sum over k equal to 1 to capital K probability of Z equal to k into probability of X given Z equal to k, which is exactly equal to sum over k equal to 1 to capital K pi k. What is x given z equal to k? x given z equal to k is normal with mean mu k and covariance matrix sigma k, we can write that as normal X with mean mu k and covariance sigma k.

So, we have not really done anything new here, the only thing we have done is that we have given a name to the cluster, our component ID, Z, that is the only thing we have done here. And this is, such a variable Z is called a hidden or a latent variable, because you never see Z in your data, you only see X. So, Z is something which is useful for modelling for us. But really, it is not available to you available to the person who is doing data analysis. So, Z is corresponds to the cluster ID.

So, in the, for example, in the exam case, students are not going to volunteer the information that be mostly are belong to type 1 or type 2 or type 3, no one is going to give that information. But it is there. So, it is there in the real world, but it is not available to you. So, that is what is called a hidden variable.

(Refer Slide Time: 16:33)

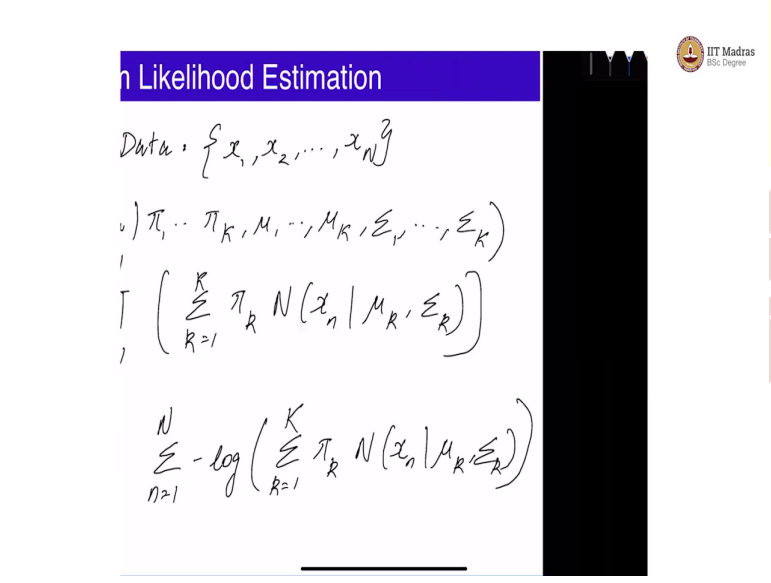


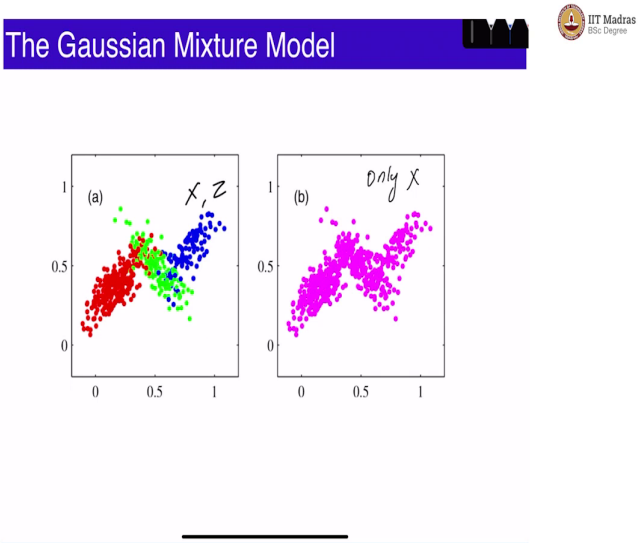
So, for example, if you, let us say, let us consider an example, three component Gaussian mixture model, if you had the real if you had the cluster, or the component ID z also, you would have data like this, that is you would have three components. And you also know which data which came from which component, so you would also have this color information, it is that corresponds to having both the data x and the cluster, or the component information set, that is a hidden variable.

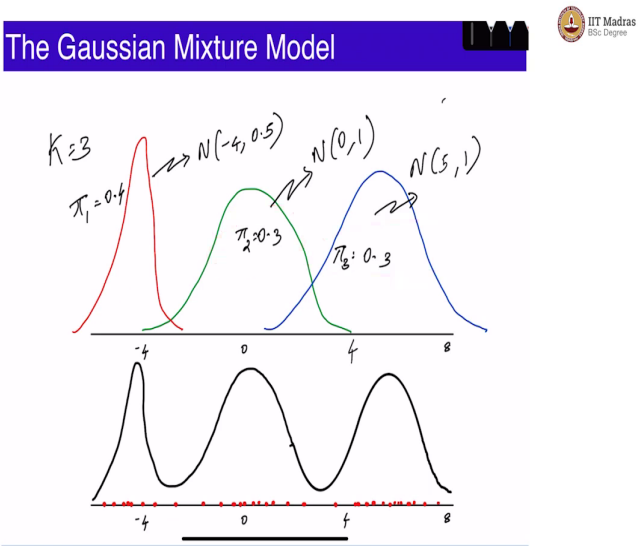
If you use it, then effectively each data point you would not only have the position, but also the color of it. But practically, you would not have that you would only have x. So, effectively, you would have data like this. So, you do not know a given point, is it from the red cluster or the green cluster, or the blue cluster, you do not know that. You only know that there is a point like this? That is what you would know.

Now you have data like this. Now, the question is, can you estimate the parameters pi k, mu k and sigma k. Because this is also a parametric family. And we have been doing parameter estimation using maximum likelihood for parametric families like normals, uniforms, normal with mean and variance and so on. Can we do the same thing for the Gaussian mixture model?

(Refer Slide Time: 17:52)







Yes, you could try. That is really maximum likelihood seems to be a valid thing to do here. And you could do that. Let us do that. First off, you should, well, before you even start, you must know the number of components, the number of components should be known a priority. So, that is, in practice, what people do is they try out different number of components. But for now, let us assume that you know the number of components that is your capital K is known to you. Let us assume that.

So, you would have data for maximum likelihood estimation, which is you have let us say n data points, x1, x2 dot dot dot xn. All of these are, let us say in D-dimensional vectors. It is a D-dimensional vector. And now you can write down the likelihood of this data given the parameters. So, you can say, what is the probability of data given you know the parameters?

What is the parameters? Well, that is pi 1 to pi k, mu 1 to mu k and sigma 1 to sigma k. These are the parameters of the Gaussian mixture model.

Well, how would you do that? Well, you have independent data, you can use the same idea that we have been using, which is independence, you can just it is a product of so product over i equal to 1 to capital N, or let us use the small index n, so we will use small n as a dummy variable, and capital N is the number of data points. What is the probability of a given data point that is the Gaussian mixture model expression, that is going to be sum over k equal to 1 to capital K pi K into normal x evaluated at xn with mean given by mu k and covariance sigma k.

You can view this entire thing as a function of mu k sigma k and pi k. Why is that, because the data points are known. So, you have n data points which are known. So, this entire thing is a function of pi k, mu k and sigma k. What do you want, you want to find the parameters phi k mu k and sigma k that maximize this term, that is a meaningful thing to ask. You want to maximize the likelihood, once again, you can do the same thing that we have been doing with maximum likelihood instead of maximizing the likelihood, we will minimize the negative log likelihood, we could do the same thing.

Well, you could do that, that would be R of let us call all the parameters together into one package, let us call it theta, R of theta as - well, that is going to be sum over n equal to 1 to capital N - log of sum over k equal to 1 to capital K pi k normal evaluated at xn with mean given by mu k, covariance given by sigma k. So, this is the expression of the negative log likelihood.

And what we want to do is you want to find the parameters pi k, mu k and sigma k, that maximizes sorry minimizes this term. You can see that this is a much harder thing than the case where a standard normal or normal distribution or uniform distribution because you have a summation inside the log, and that is always ugly. So, if you have product inside the log that is nice, because you can simplify it.

But if you have summation inside a log, that is a ugly thing, and you just cannot directly minimize this function, you could find the gradient of it, but you cannot set it to 0 easily. So, you cannot set it to 0 and solve for it. And that is a hard thing to do. So, even though we have an expression for the negative log likelihood, minimizing this is a hard thing to do. You could use generic optimization procedures, like gradient descent, things like that. But let us say we

are not satisfied with that we want a more intuitive solution. Let us say we want a more intuitive solution. How do you go about doing that?

Let us take a step back and think about this first. We have data like this. Somehow, let us say, magically, let us say we had access to data like this. If we had access to data like this, then finding the parameters pi k, mu k and sigma k is pretty easy. How do you do that? Well, to get the mu corresponding to the red, you simply take the sample mean of the red data points.

To get the covariance corresponding to the red that is sigma. To get sigma red, you simply find the sample covariance of the red points. To get the pi, the fraction of the pi parameter corresponding to red, you simply compute the fraction of red points, so let us say you find one-third of the points are red. So, it means that pi red will be equal to 1 by 3, it means that if you had access to which data point came from which cluster or which component, then parameter estimation is ridiculously easy, you just have to just do a sample average, and you can get the parameters. That is the first point to note.

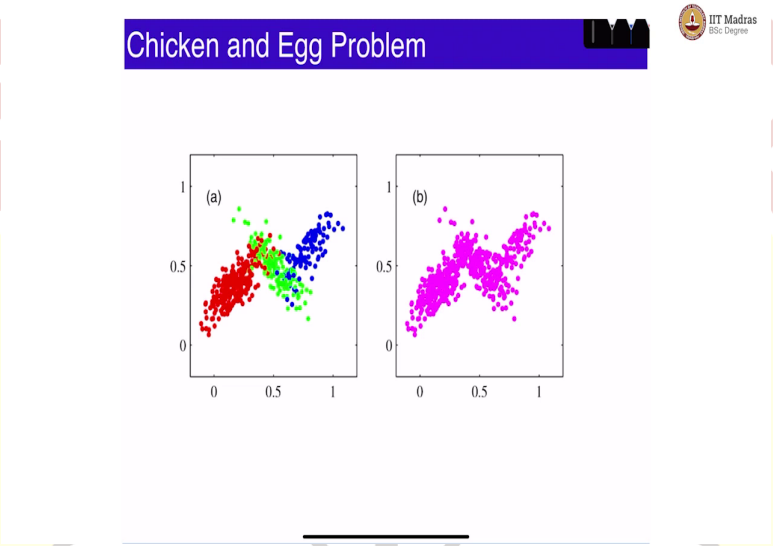
The second point to note is also interesting. That is, if you had somehow magically had access to the parameters, let us say if you somehow magically had access to the parameters, then finding which data point belong to which component is also somewhat easy. Let us take the simple one-dimensional example here. Let us say you somehow knew the parameters. Let us say you knew the parameters pi, pi 1, mu 1, sigma 1; pi 2, mu 2, sigma 2; and pi 3, mu 3, sigma 3.

Let us say you knew these parameters, and someone gave you the data point, let us say 0.3. You should, or someone gives 0.0.2, let us say someone gave a data point 0.2, you immediately know for almost certainly that this data point 0.2 came from the green cluster, the green, the second component. You, knowing the parameters also helps you to identify which component, that particular data point came from. This is not always so straightforward.

For example, if you have a point like 0.2, you know that it came from the green point, if you have a point, let us say 6.2, you know that it came from the blue component, if you have a point like let us say - 5.3, you know, it came from the red component, but what about the point like this, let us say you have a point like - 2. A point like - 2 could have come from the red cluster, the red component, it could have also come from the green component. So, a point like - 2 can come could have come from either of the two components.

So, it is for some data points, it is not possible to assign that data point to one single component. But still, you can, you can do something about it. At least for a large number of data points, you can assign a component to each of the data points. So, put it together, what we have is that if you had access to the red, the parameter, the cluster or the component IDs, you can easily compute the parameters. And if you had the parameters, you could sort of compute the component IDs. But unfortunately, you have neither.

(Refer Slide Time: 25:26)



So, if you have the parameters, you can compute the component IDs, if you have the component IDs, you can compute the parameters, but you have neither. So, that is what is called a chicken and egg problem. So, if you have a chicken, you can get an egg. If you have an egg, you can get a chicken, but you have neither you do not have a chicken or an egg. But somehow if you could somehow get one of them, then you can get the other from it. So, how do you solve this kind of problem?

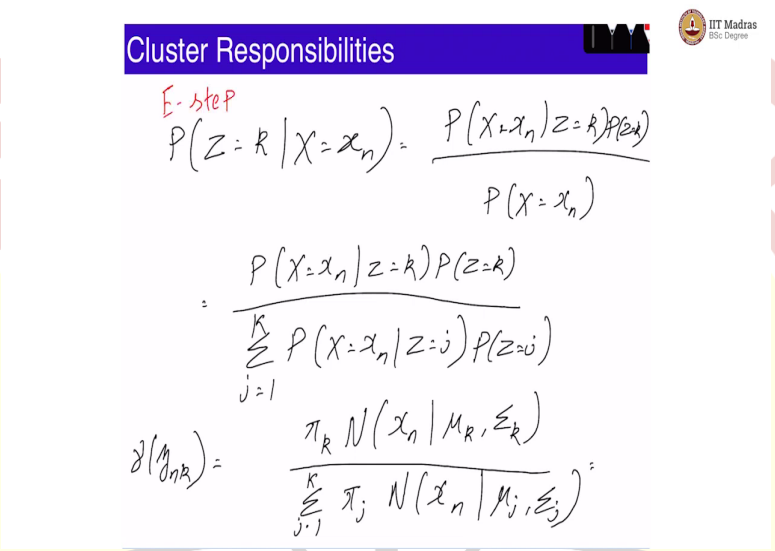
So, this kind of problem is typically this is very often encountered in machine learning. And the typical way to solve it is first create a random chicken. So, you create a random chicken first, and then you create an egg from the chicken. And then you create chicken from that egg. And you keep on adding generations, and you hope that you get a chicken and an egg. So, this is an analogy. If you want to do the same thing, how would you go about doing that?

Well, if you had the parameters, I said that you can compute the which data, which component does each point belong to, you can compute that. So, let us just create give a random parameters. So, let us just initialize the parameters to some random numbers. Let us

say we set pi 1, pi 2, pi 3, mu 1, mu 2, mu 3, sigma 1, sigma 2, sigma 3 to some valid numbers, you, now you have set those numbers.

Once you have set those numbers, can we compute which component does each data point belong to? I give an intuitive idea for that in the one dimensional example. But how do we generalize the idea?

(Refer Slide Time: 26:51)



Well, a standard easy way to mean standard way to do that might be as follows, which is you let us say now you have the parameters, let us say now you are fixing the parameters. Now you have pi 1, pi 2, pi k, mu 1, mu 2, mu k, sigma 1, sigma 2, sigma k, you have that which means that you have the joint density of x , z. With that, if you want to compute this, what is the probability of Z equal to k given let us say x is equal to xn? You can ask this question. So, how to interpret this?

Well, for the nth data point, what is the probability that the cluster or the component it belongs to its k? That is the question I am asking. How do I answer this question? Well, you can use Bayes rule to say this is equal to probability of X equal to xn given z equal to k into probability of z equal to k divided by probability of X equal to xn. So, the denominator you get by summing over j, let us simplify that even further. This is probability of X equal to xn given z equal to k into probability of z equal to k divided by sum over j equal to 1 to capital K probability of X equal to xn given z equal to j into probability of Z equal to j.

This is exactly the Bayes rule and why is this useful because we know that x given z it is simply a Gaussian with mean mu k and covariance sigma k and probability of Z equal to k is

simply equal to pi k. So, with that we can write this as simply pi k normal evaluated at xn with the mean given by mu k and sigma k and the denominator is simply given by sum over j equal to 1 to capital K pi j normal evaluated at xn with mean given by mu j and sigma j.

So, this is a natural notion where you assign each data point to you say you find what is the probability that data point xn belongs to cluster K. You can easily see that let us denote this. Let us denote this object by gamma of Z nk let us need more space. Let us denote this object by gamma of Z nk.

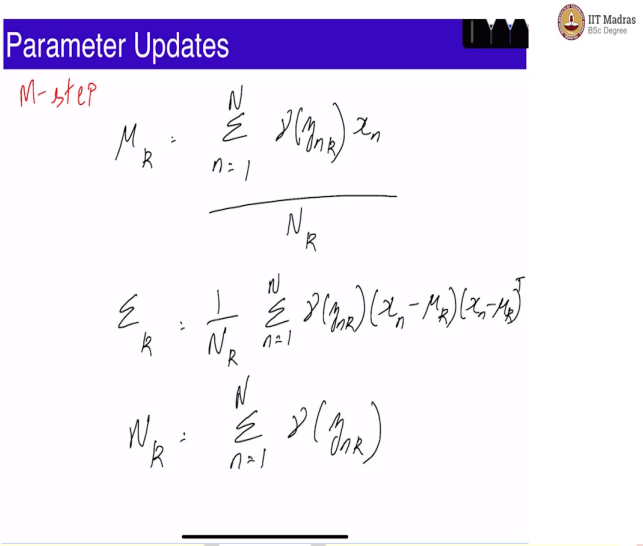
So, the interpretation of gamma of Z nk is what fraction of our mean what is the probability that data point n belongs to cluster K. That is the interpretation of gamma of Z nk. Clearly, if you sum over k equal to 1 to capital K gamma of Z nk you get 1. Why is that, because you sum over k equal to under on the capital K, that is exactly the numerator and denominator are going to match. So, that is going to be equal to 1.

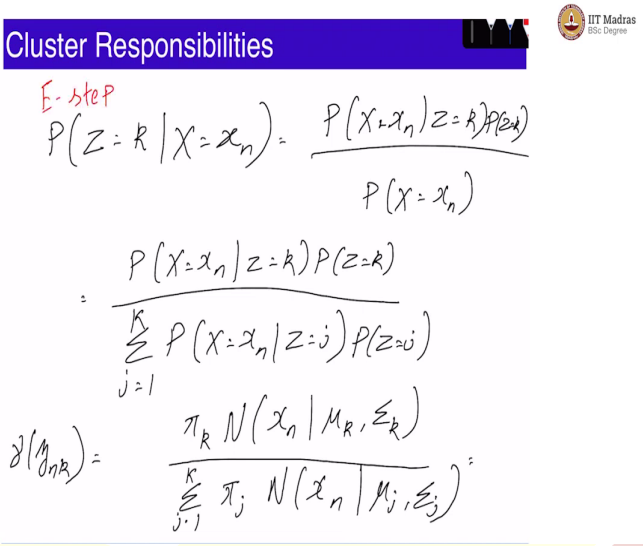
So if gamma, so now this is called cluster responsibilities. So, that means that if you have the parameters, pi 1 to pi k, and mu 1 to mu k and sigma 1 to sigma k, you can find cluster responsibilities. That is, you can find a way to assign a cluster to each data point. Ideally, you will find one cluster will take full responsibility for data point by that I mean gamma of Z nk will be either 1 or 0. Let us take that simple case first.

Technically, gamma of Z nk can be any number between 0 and 1. But it has to sum to 1. If you sum over k equal 1 to capital K gamma of Z nk should be equal to 1, but it can be any number between 0 and 1. Let us for simplicity sake, let us say gamma of z nk is exactly equal to 1 for just one k. And for every other k it is 0. That means that every data point is assigned to just one cluster? So, let us take the extreme case.

In that scenario, how would you compute the parameters mu and sigma. So, now we created a random chicken which was the randomly created parameters pi, mu and sigma and from that we assigned a component in each data point. Now we have created from the random chicken, we have create an egg, which is the Z, the gamma of Z nk. And now from this egg, can we hatch a new chicken. And the answer is, you can do that, how would you do that? Well, you just consider all the data points assigned to cluster 1 and you take the sample mean from that would correspond to mu k.

(Refer Slide Time: 31:46)





So, mu k would simply the sum over n such that gamma of Z nk is equal to 1 xn divided by number of data points assigned to cluster k, that is cardinality of the set of points such that cardinality of n such that gamma of Z nk is equal to 1. So, now, let us simplify this, instead of having gamma of Z nk just assuming 0 or 1, we can just simplify this as follows, you can just say sum over n equal to sorry, sum over some more n equal to 1 to capital N, gamma of Z nk times xn.

So, in the situation where gamma of Z nk can take only values 1 or 0; gamma of Z nk will be equal to 1, if data point xn is assigned to cluster k, it will be 0 otherwise. So, effectively by doing this, I have summed over all points that are assigned to cluster k. And I can just divide that by the number of points assigned to cluster k. So, let us call that Nk. And what will Nk

be? Nk will simply be equal to the number of points assigned to cluster K, which will simply be sum over n equal to 1 to capital N, gamma of Z nk. So, this is a proxy for the number of points assigned to cluster k.

So, I just took the points assigned to cluster k, and it took a sample mean, and that is my parameter mu k. And you can do the exact same thing for the covariance also. Now, my sigma k will simply be 1 by Nk, sum over n equal to 1 to capital N, gamma of Z nk times xn - mu k into xn - mu k transpose. So, the mu k is the same as this. So, xn - so this is the sample covariance.

So, I just compute the sample covariance of the data points assigned to cluster k. That is why I have a gamma of Z nk. Gamma of Z nk will be exactly equal to 1 if data point n is assigned to cluster k, and it will be 0 otherwise. So, in that situation, this will exactly correspond to the sample covariance of the data points assigned to cluster k. But, of course, gamma k is not going to be exactly 0 or 1, it is going to be between 0 and 1. But that does not stop you.

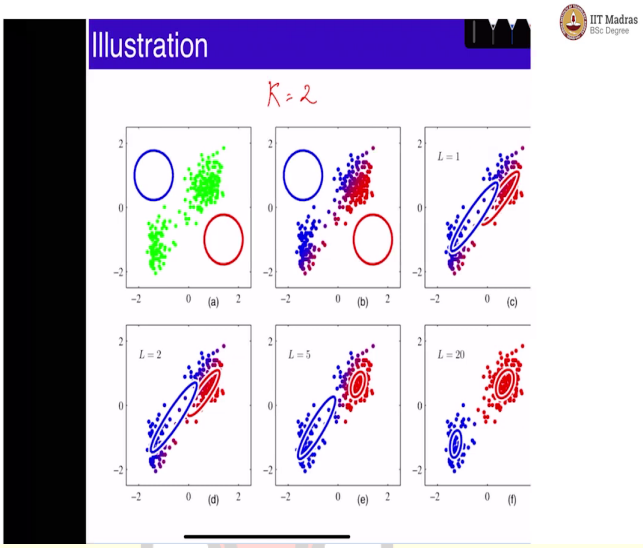
This is still interpretable in that way, gamma even if gamma of 0 is not exactly 0 or 1, you can this expression is still meaningful and can be interpreted in that way. So, let us, if you are finding it difficult, just let us take gamma of Z nk will always take either values 1 or 0. In that case is pretty easy to interpret, even if not, you can apply this expression. This is the parameter update.

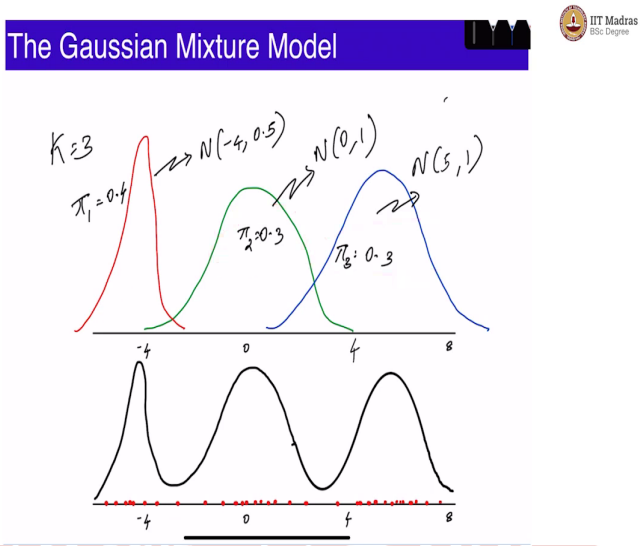
So, now from the egg that you hatched from the random chicken, you have created a new chicken. Hopefully, this chicken is more realistic than the random chicken. That is the hope. So, now you have a new chicken. And now you can create from this new chicken, you can create a new egg, which is hopefully more realistic than the previous egg. And you can create a new chicken and so on, you can keep on looping this. And eventually, the hope is that you will end up getting a very realistic chicken and egg.

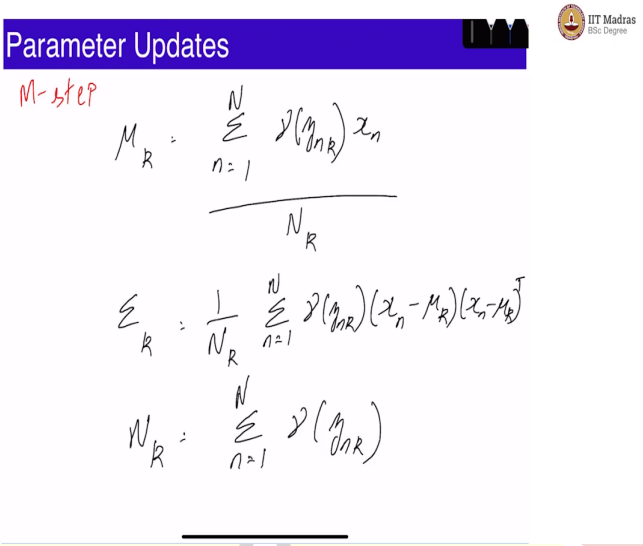
So, this is analogy, and I am not going to explain these expressions more in more detail than this. So, but we can do that in a later course on machine learning. But intuitively, that is what is happening here in these updates. So, these updates, this thing of finding the cluster responsibilities, when you are given a parameter is called the E-step. And finding the parameters from when you know the cluster responsibilities come opposite, and k is called the M-Step.

Put these together, you get what is called the EM algorithm, you first initialize the parameters randomly you do the E-step, then the M-step, then the E-step, then the M-step, and so on, you can keep on alternating till you have reached convergence.

(Refer Slide Time: 36:38)







So, let us have a very simple illustration of what happens. Let us say you have data like this, we can see that the data clearly has two components. Let us say you happen to know capital K equal to 2. So, let us say you know capital K equal to 2. You initialize two components mu k and sigma k this way like so, you initialize mu k to be this and sigma k corresponding to a circular covariance, let us say sigma k is initialized to the identity covariance. So, you might end up with this is a contour plot of the other two densities of the components.

So, this is the normal the, if you draw a contour plot of normal with mean mu k and covariance sigma k, you get this and sorry, mu 1 and sigma, new blue and sigma blue, you get this. And you if you do the contour plot for one of the contour plots for normal with mean mu to mu red and sigma red, you get this. So, initialized, you created your random chicken and once you have these things, now you can compute your gamma of Z nk's. Gamma of Z nk in this case, it is just 2, you mean there are only two possibilities, there are only two components, either blue or red.

So, gamma of Z and blue and gamma of Z and red. So, that is what you have. And you can kind of see that is what is happening here. If you have a point, which is mostly red, you can see it is red, if you have a point which is mostly blue it is blue, but there are points somewhere here, which are halfway between blue and red. So, that is why you have these points are mostly red, these points are mostly blue, because these points are closer to the blue component, these points are closer to the red component.

May be I will use a different coloured pointer, I will use from green point. So, these, these points are closer to the red component. So, these points are assigned the red color. And these

points are closer to the blue component, these are assigned to the blue color. You have points in the middle, which are kind of equidistant between blue and red. So, you can see that it is kind of a purple color coming here that corresponds to let us say, gamma of Z and blue is equal to gamma of Z and red equal to let us say something like 0.5.

So, it is possible for gamma of Z and blue and Z and red to be equal to 0.5. So, that is given by k. It does not matter; you can have that such a thing. And now you have your cluster responsibilities, you can compute, recompute the parameters using the parameter which is the M-step. So, this is the E-step, you got the cluster responsibilities, and once you got the cluster responsibilities gamma of Z nk, you can compute the M-step which is you compute the sample mean and the sample covariance of the blue points and the red points.

So, if you do that, you get something like this you can see that the blue points are mostly here, and read points are mostly here and that is reflected in this thing you can see a blue points here red points here. So, now you have computed, now you have a different mean and means and covariances you can recompute the cluster responsibilities.

So, you can see that you can compute that. Again, you can do the E-step again. And you can see that the colors now change, you can see that these points are red, these points are blue, you can now, now that you have computed the E-step again, now we can do the M-step again now and you can recall you will get the mean and the covariance of the blue and red component changes so, it becomes this and this.

And you can keep on doing it. Let us say you did for 20 steps. After 20 steps, you end up with the blue component like this and the red component like this, which is pretty good, even though you started with something random, you ended up with two components, which do capture the two modes of the data. So, if you had the data like this, data is clearly bimodal. So, you have a peak here, and you have a peak here, and you have pretty much very low density outside it.

And your expectation maximization algorithm, which is the EM algorithm, really did figure out these two components. So, this is how you would apply the EM algorithm on Gaussian mixture model data. This is how you use the EM algorithm to estimate parameters of the Gaussian mixture model.

So, the Gaussian mixture model as a very, it is a fundamental building block of a lot of machine learning even today, and it is a fundamental building block for a pre-processing step,

which is called clustering. So, you have, let us say, let us even take the score example that examine the test example that we discussed in the beginning, you have students clustering to three types of students.

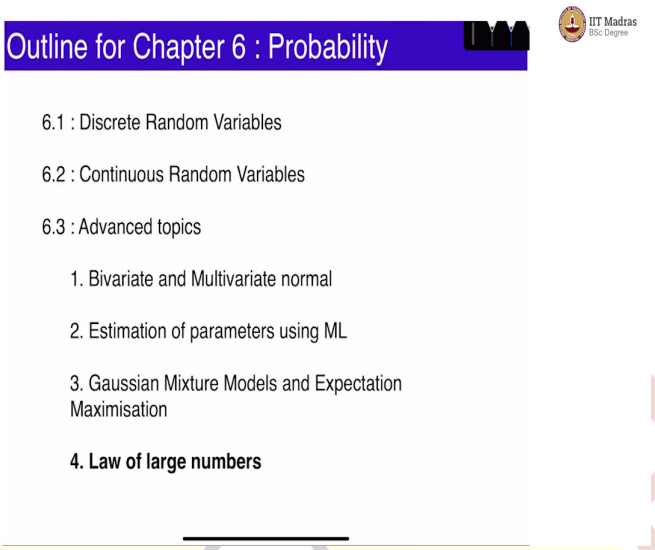
So, if you wanted to group the students into three types, or you would run an EM algorithm on the score data, if you could compute the scores, and you could run the EM algorithm, you will figure out that there are three types of people. For one-dimension, it is not too hard, you do not need algorithms, you can just plot the histogram and figure it out. But what if it is not one-dimensional data, if it is, let us say 10-dimensional data, it is no longer possible to figure it out using just the, for one-dimension, you could have just like drawn this joint density.

The density here, and then figured out that there are three modes; mode 1, mode 2, mode 3 and we are done with it. But if you have 10 dimensional data, or even 100 dimensional data, you can no longer visualize the data easily. So, you have to, you want to cluster the data into let us say multiple modes, the standard approach is to use the EM algorithm to learn to estimate the parameters of a Gaussian mixture model.

So, Gaussian mixture model is one of the building blocks of clustering and machine learning. And this is simply a classic algorithm that is in the must know category of every machine learning student. In a more detail machine learning course, you will see the theory behind the why these updates, not only are intuitively nice, they also make theoretical sense.

So this, right now I have just given intuitive idea behind why these updates are meaningful, but not only they are intuitively meaningful, they are also theoretically well founded. So we will in a later machine learning course, you will see what that is the case. With that we can wrap up the discussion on Gaussian mixture models.

(Refer Slide Time: 42:44)



And we will move on to the next topic and next and final topic in this chapter.